27 While useful, the rubber-band analogy is not perfect. The inward, negative pressure exerted by the rubber bands impedes the expansion of the box, whereas the inflaton's negative pressure drives the expansion of space. This important difference illustrates the clarification emphasized on this page: in cosmology, it is not that uniform negative pressure drives expansion (only pressure differences result in forces, so uniform pressure, whether positive or negative, exerts *no* force). Rather, pressure, like mass, gives rise to a gravitational force. And negative pressure gives rise to a repulsive gravitational force that drives expansion. This does not affect our conclusions.

28 Some researchers, including Alan Guth and Eddie Farhi, have investigated whether one might, hypothetically, create a new universe in the laboratory by synthesizing a nugget of inflaton field. Beyond the fact that we still don't have direct experimental verification that there is such a thing as an inflaton field, note that the twenty pounds of inflaton field would need to be crammed in a tiny space, roughly 10^{-26} or so centimeters on a side, and hence the density would be enormous—some 10^{67} times the density of an atomic nucleus—way beyond what we can produce, now or perhaps ever.

29 Don't get confused here: The inflationary stretching of quantum jitters discussed in the last section still produced a minuscule, unavoidable nonuniformity of about 1 part in 100,000. But that tiny nonuniformity overlaid an otherwise smooth universe. We are now describing how the latter—the underlying smooth uniformity—came to be.

30 For ease of writing, we'll consider only fields that reach their lowest energy when their values are zero. The discussion for other fields —Higgs fields—is identical, except the jitters fluctuate about the field's *nonzero*, lowest-energy value. If you are tempted to say that a region of space is empty only if there is no matter present and all fields are *absent*, not just that they have the value zero, see notes section.[2]

31 The remainder of this chapter recounts the discovery of superstring theory and discusses the theory's essential ideas regarding unification and the structure of spacetime. Readers of *The Elegant Universe*